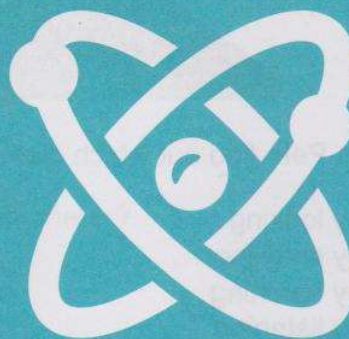




INTRODUCTORY PAPER



2018 SCIENCE

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. We use our senses to learn about things around us.

Pete has two glasses of water. One glass is filled with warm water and the other is filled with cold water.



How can Pete find out which glass has the warm water?

- (A) by looking
- (B) by smelling
- (C) by touching
- (D) by listening

2. On Sunday, Frank checks the weather chart for the next five days.

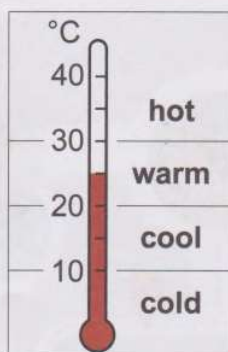
Monday	Tuesday	Wednesday	Thursday	Friday
				

On how many of these days can Frank expect it to rain?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

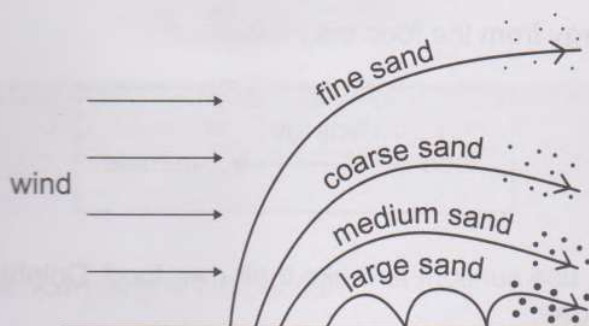
For questions 3 and 4 use the information below.

Ms Thompson has a thermometer hanging over a chart in her Year 2 classroom.



3. Why does Ms Thompson place the chart with the thermometer?
- (A) to keep the thermometer warm
 - (B) to measure the length of the thermometer
 - (C) to show what temperature it is when you feel cold or warm
 - (D) to measure how long it takes for the classroom to become warm
4. What is the temperature reading on the thermometer?
- (A) 20 °C
 - (B) 25 °C
 - (C) 30 °C
 - (D) 35 °C

5. This diagram shows the effect of wind on sand grains of different sizes.



What size of sand grains are likely to be moved farthest by the wind?

- (A) fine sand
- (B) coarse sand
- (C) medium sand
- (D) large sand

6. The map shows a zoo that Year 2 students will visit on an excursion.

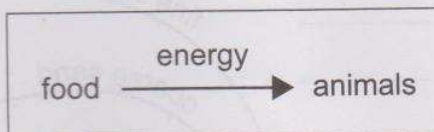
The students will follow the dotted line to go around the zoo.



How many different animals will the students stop to see during the visit?

- (A) 5 (B) 7 (C) 8 (D) 10

7. Animals get their energy from the food they eat.



Plants such as algae, use sunlight to make their own food. Dolphins eat fish. Fish eat shrimp. Shrimp eat algae.

Which of the following correctly shows how energy moves through the food chain of these living things?

- (A) dolphins → fish → shrimp → algae
(B) fish → dolphins → shrimp → algae
(C) sunlight → fish → algae → dolphins
(D) algae → shrimp → fish → dolphins

8. Things around us change. Some changes can be easily undone and others cannot.

Which of these changes cannot be undone?



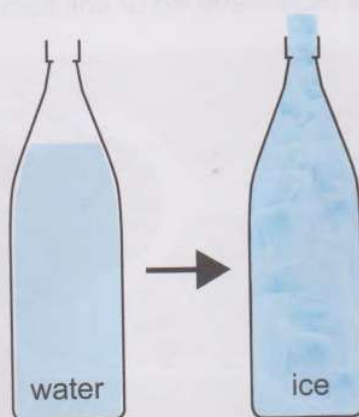
9. Luisa has this pebble.



How many centimetres (cm) long is the pebble?

- (A) 2
- (B) 5
- (C) 7
- (D) 9

10. Jason placed a bottle of water in the freezer. When he took the bottle of water out of the freezer the next morning, this is what he saw.

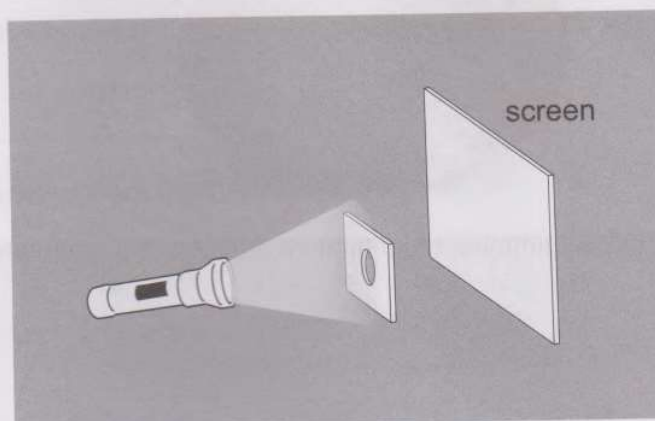


What is the best explanation for this?

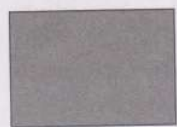
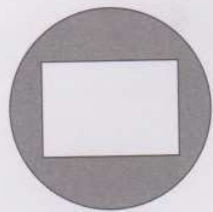
- (A) Water takes up more space when frozen.
- (B) Water becomes heavier when frozen.
- (C) Water bubbles up when freezing.
- (D) Water spills before freezing.

11. Tina used her torch to make shadows on a screen. She had a piece of cardboard with a hole cut out of the middle.

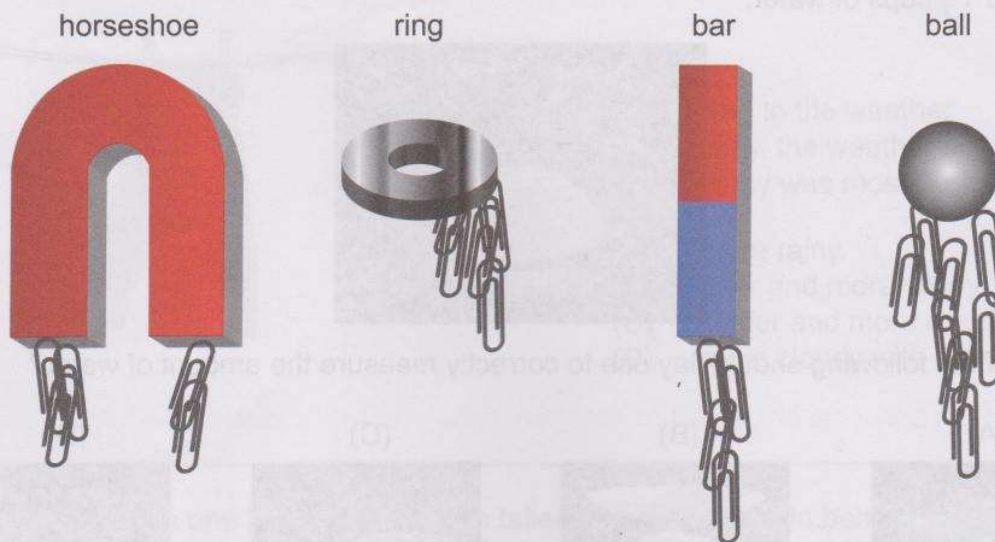
Tina shone the light from the torch onto the cardboard as shown in the diagram.



What did the shadow on the screen look like?

- (A)  (B)  (C)  (D) 

12. Ned has four different magnets. He knows that magnets can pick up small iron objects. He used each magnet to pick up as many iron paperclips as it could.



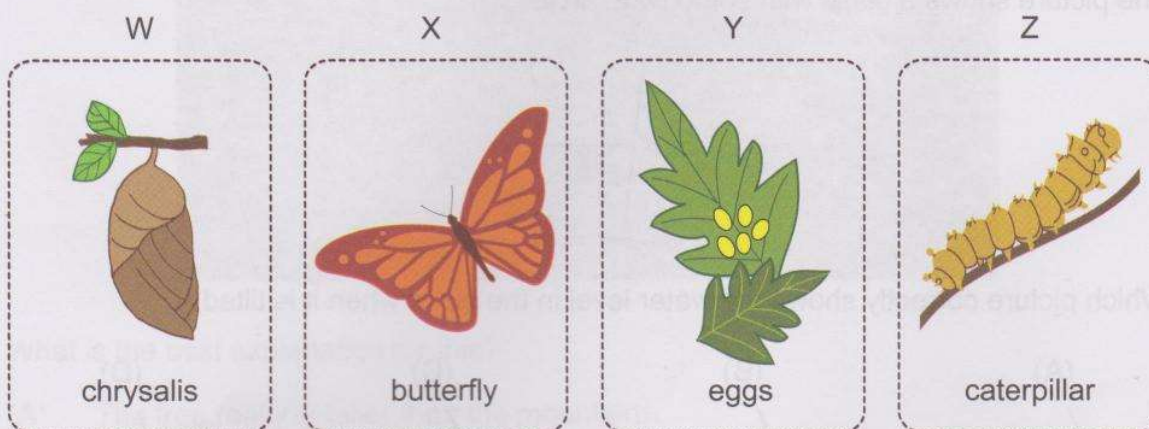
Which is the strongest magnet?

- (A) ball (B) bar (C) ring (D) horseshoe

13. There are four stages in the life cycle of a butterfly:

- Stage 1: A female butterfly lays her eggs on plant leaves.
- Stage 2: A caterpillar is hatched from each egg.
- Stage 3: The caterpillar becomes a chrysalis.
- Stage 4: An adult butterfly comes out of the chrysalis.

These cards show the four stages, but not in the correct order.



Which of these represents the correct order of stages for the life cycle of a butterfly?

- (A) W → X → Y → Z (B) X → Z → Y → W
(C) Y → Z → W → X (D) Y → W → Z → X

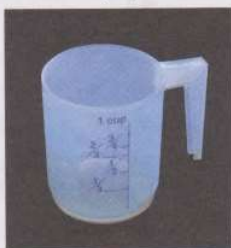
For questions 14 and 15 use the information below.

Jay's mum is making some dough. She has some flour in a bowl. She tells Jay that she will need $1\frac{1}{2}$ cups of water.



14. Which of the following should Jay use to correctly measure the amount of water?

(A)



(B)



(C)



(D)



15. Jay's mum then decides to double the amount of flour in the bowl to make more dough. How many cups of water will she need altogether to make the dough?

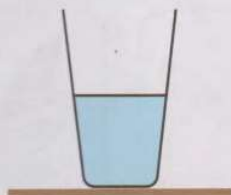
(A) $1\frac{1}{2}$

(B) 2

(C) $2\frac{1}{2}$

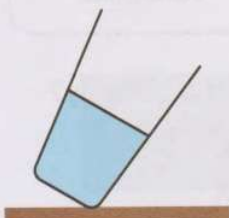
(D) 3

16. The picture shows a glass with some water in it.

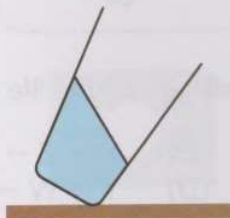


Which picture correctly shows the water level in the glass when it is tilted?

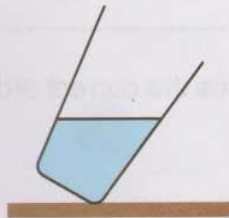
(A)



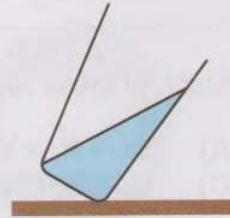
(B)



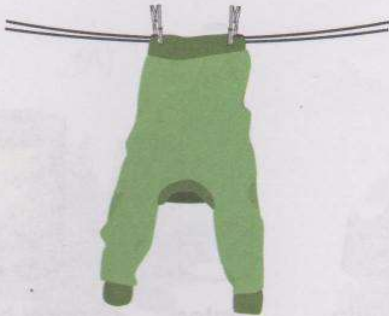
(C)



(D)



17. On Monday morning Joe washed his sweatshirt and hung it outside. It took 4 hours to dry. On Wednesday morning Joe washed the same sweatshirt again and hung it outside. It took 2 hours to dry.



Compared to the weather on Monday, the weather on Wednesday was most likely

- (A) more rainy.
- (B) drier and more windy.
- (C) colder and more cloudy.
- (D) more cloudy with no wind.

18. In this photograph one tree appears to be taller than the mountain behind it.

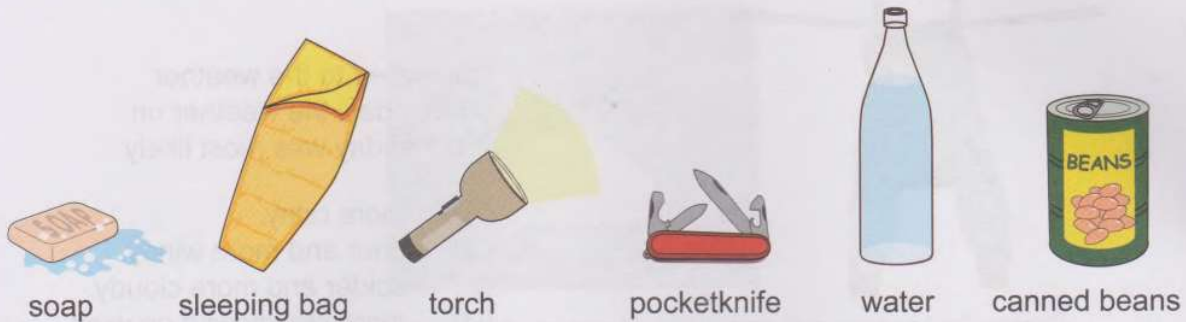


What is the best explanation for this?

- (A) The tree really is taller than the mountain.
- (B) The tree is closer to the camera than the mountain is.
- (C) The tree must be standing on an even higher mountain.
- (D) The camera that took the photograph must not have worked properly.

19. People need to breathe air, drink water, eat food and keep warm to stay alive.

Jen is going camping in the mountains for three days. She is preparing things to take with her from these:



Which of the following are the three most important things for her to take?

- (A) sleeping bag, torch, water
- (B) torch, pocketknife, sleeping bag
- (C) soap, sleeping bag, water
- (D) sleeping bag, water, canned beans

20. Lily is Henry's older sister. Lily weighs 25 kilograms.

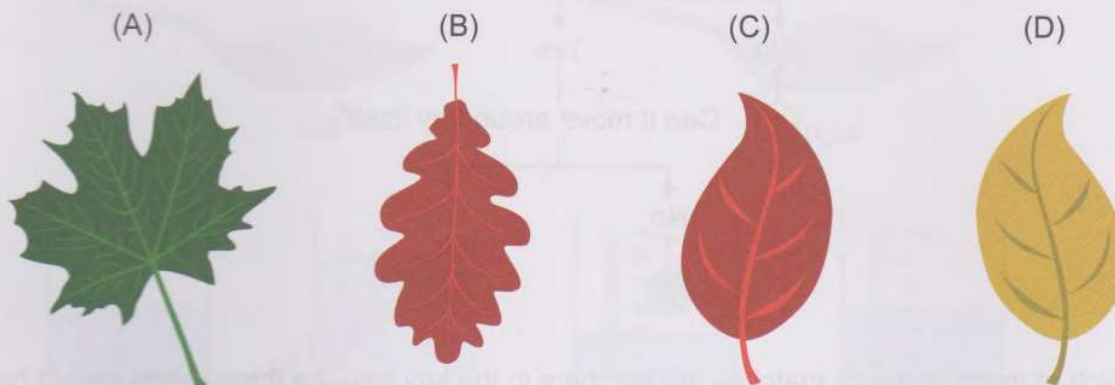


How heavy is Henry most likely to be?

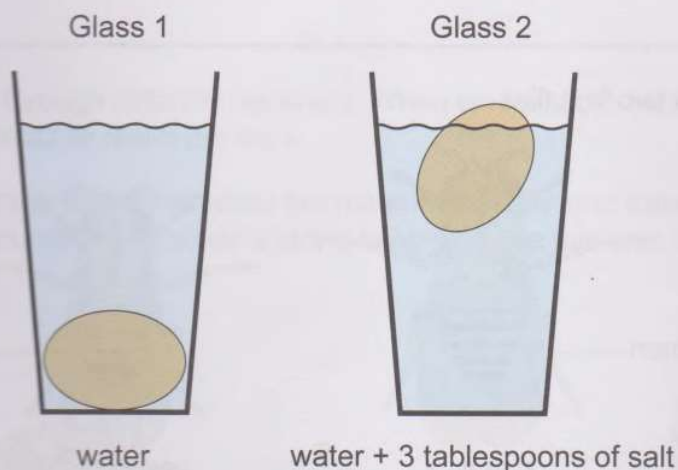
- (A) 5 kilograms
- (B) 10 kilograms
- (C) 20 kilograms
- (D) 50 kilograms

21. Some leaves have the same colour but different shapes. Some leaves have the same shape but different colours.

Which leaf is not like any of the others in shape or colour?



22. Mr Martin placed a fresh, raw egg into a glass of water. It sank to the bottom. He added 3 tablespoons of salt to a second glass of water and stirred it well. When he put the same egg into the salty water, it floated.

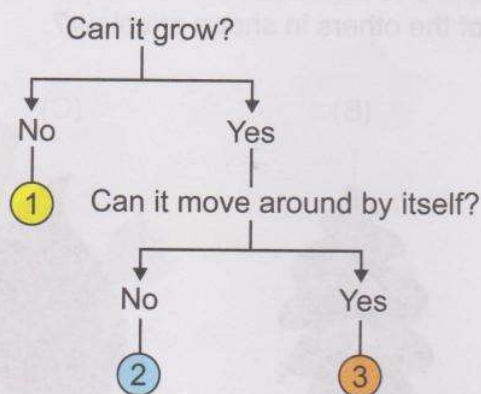


Mr Martin made another glass of salty water, using 1 tablespoon of salt.

Where would the egg settle in this glass of salty water?

- (A) the same as it did in Glass 1
- (B) higher than it did in Glass 2
- (C) the same as it did in Glass 2
- (D) between where it did in Glass 1 and in Glass 2

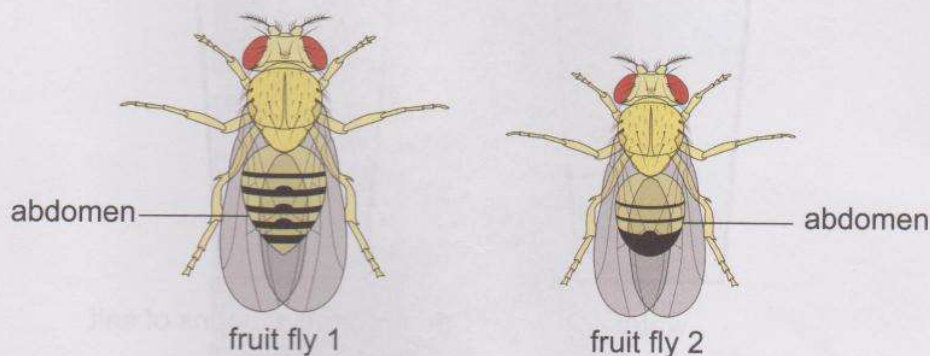
23. Jessie had a plant, a cat and a toy car. Her dad drew this key to show her the differences between the three things.



Which of these correctly matches the numbers in the key with the three things Jessie had?

	1	2	3
(A)	toy car	cat	plant
(B)	plant	cat	toy car
(C)	toy car	plant	cat
(D)	cat	toy car	plant

24. This picture shows two fruit flies.



There are differences between the body of a male fruit fly and that of a female fruit fly.

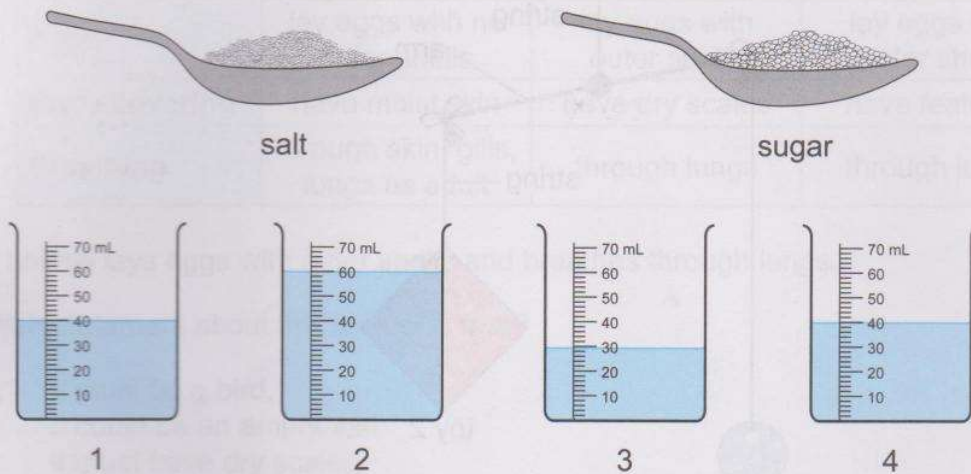
- Males have a round abdomen, females have a pointed abdomen.
- Females are usually larger than males.
- The last section of the abdomen in a male is darker than that of a female.

Which sentence about these two fruit flies is correct?

- (A) They are both male.
 (B) They are both female.
 (C) Fruit fly 1 is a male and fruit fly 2 is a female.
 (D) Fruit fly 1 is a female and fruit fly 2 is a male.

25. Eve wants to see which dissolves faster in water, salt or sugar.

She takes a teaspoon of salt and a teaspoon of sugar. Both teaspoons are the same size. Each amount of salt and sugar is the same.



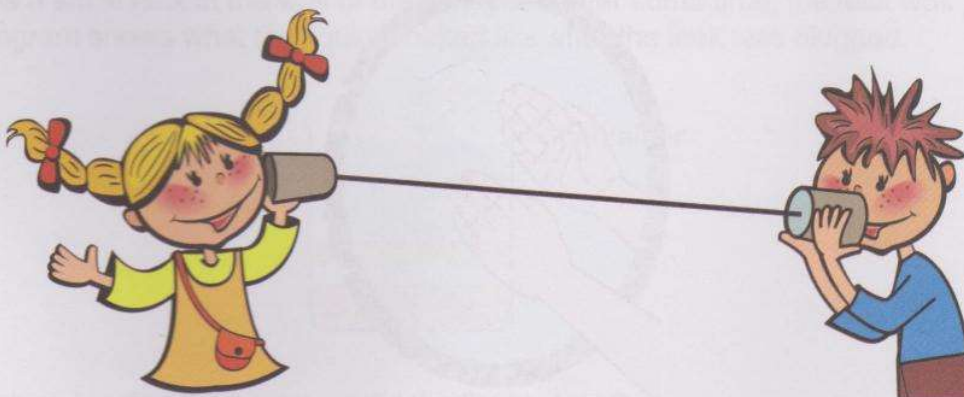
She has four containers of water, but wants to use two containers.

Which two containers of water must Eve use?

- (A) 1 and 2 (B) 2 and 3 (C) 3 and 4 (D) 1 and 4

26. Sound can travel through different materials. When we speak to each other, the sound travels through the air to reach our ears.

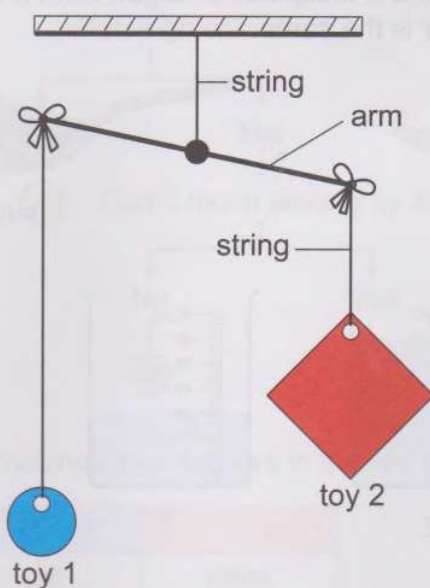
Jill and David want to find out whether the material that a sound travels through has an effect on how it sounds. They made a string telephone like this one, using a cotton string and two cups.



What else do they need for their investigation?

- (A) a real telephone
(B) strings made of other materials
(C) cotton strings of different lengths
(D) other people to listen to the telephone

27. Liam's baby sister has a mobile hanging over her cot. The lengths of the two arms are the same. One day Liam noticed that it was not balanced.



Which of following is the most likely reason why the mobile is not balanced?

- (A) The arm is too short.
- (B) One string is too long.
- (C) Toy 1 is heavier than toy 2.
- (D) Toy 2 is heavier than toy 1.

28. When you see yourself in a mirror, you appear to be reversed from left to right. For example, when you see your right hand in the mirror, it looks like your left hand.



Jenny wrote 'ME' on her hand and held it facing a mirror.

What did the word look like in the mirror?

- (A) EM
- (B) EM
- (C) ME
- (D) WE

29. The table shows some information about three different groups of animals: amphibians, reptiles and birds.

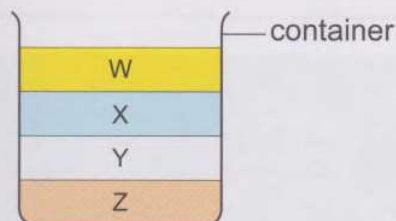
	Amphibians	Reptiles	Birds
Eggs	lay eggs with no outer shells	lay eggs with outer shells	lay eggs with outer shells
Body covering	have moist skin	have dry scales	have feathers
Breathing	through skin, gills, lungs as adult	through lungs	through lungs

An animal lays eggs with outer shells and breathes through lungs.

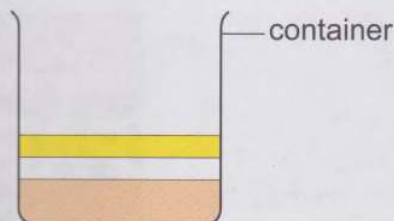
Which statement about this animal is true?

- (A) It must be a bird.
- (B) It could be an amphibian.
- (C) It must have dry scales.
- (D) It could be a reptile or a bird.

30. The diagram shows four layers of different liquids in a container. The liquids do not mix with each other.



There is a small leak in the wall of the container. After some time, the leak was plugged. The diagram shows what the liquids looked like after the leak was plugged.



From which layer in the container did the liquids leak?

- (A) W
- (B) X
- (C) Y
- (D) Z

ANALYSIS BY QUESTION

The following table shows the questions you answered correctly ☒ and shades the questions that you answered incorrectly ☐. The questions are ranked from hardest at the top to easiest at the bottom.

	Question content	Area assessed	Question number	Correct answer	Your answer	Students correct in AU
Difficult Questions	Summarise relationships between organisms using a food chain	Interpreting	7	D		8%
	Use reasoning to infer the cause of a phenomenon	Reasoning/Problem solving	30	C		9%
	Calculate a total volume in given scenario	Observing/Measuring	15	D		21%
	Use reasoning to infer the cause of a phenomenon	Reasoning/Problem solving	27	D		25%
	Predict a simple effect caused by a given change	Predicting/Concluding	16	C		28%
	Interpret a map	Interpreting	6	B		30%
	Draw a conclusion based on a simple flow chart	Predicting/Concluding	23	C		32%
	Identify the equipment needed for an investigation	Investigating	26	B		36%
	Estimate a value of mass based on visual comparison	Observing/Measuring	20	C		40%
	Use reasoning to predict the outcome of a test	Predicting/Concluding	28	A		41%
	Draw a conclusion based on information in a table	Reasoning/Problem solving	29	D		44%
	Measure the length of an object using a ruler provided	Observing/Measuring	9	B		48%
	Use reasoning to infer the cause of a phenomenon	Reasoning/Problem solving	10	A		51%
	Recognise different types of changes	Interpreting	8	C		56%
	Predict the shape of a shadow cast by an object	Predicting/Concluding	11	B		57%
Easy Questions	Use information provided to make a prediction	Predicting/Concluding	22	D		58%
	Predict weather conditions based on data	Predicting/Concluding	17	B		58%
	Distinguish male and female animals based on the features in a diagram	Interpreting	24	D		60%
	Choose a controlled variable in an investigation	Investigating	25	D		63%
	Identify the purpose of a piece of equipment	Investigating	3	C		64%
	Identify the items essential for a given excursion	Interpreting	19	D		65%
	Use reasoning to infer the cause of a phenomenon	Reasoning/Problem solving	18	B		68%
	Read a temperature measurement	Observing/Measuring	4	B		69%
	Measure magnetic strength	Predicting/Concluding	12	A		71%
	Sequence the stages in the life cycle of a living organism	Interpreting	13	C		73%
	Identify a suitable tool for a given measurement	Observing/Measuring	14	A		75%
	Draw a conclusion from information in a diagram	Predicting/Concluding	5	A		76%
	Identify the odd one out in a set of objects	Reasoning/Problem solving	21	A		79%
	Interpret a weather chart	Interpreting	2	B		82%
	Identify the sense associated with a particular observation	Observing/Measuring	1	C		88%

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SOURCES

Question 8 Sand castle image by Amanda B Bones https://commons.wikimedia.org/wiki/File:Sand_castle_on_Harbour_Island_2011.jpg licensed under the Creative Commons Attribution 2.0 Generic license <https://creativecommons.org/licenses/by/2.0/>.

PRACTICE QUESTION

Which of these animals lays eggs?

- (A) dog
- (B) cat
- (C) horse
- (D) chicken

Turn your answer sheet over and fill in the correct oval.

THE FOLLOWING YEAR LEVELS SHOULD SIT THIS PAPER:

Australia	Year 2
Brunei	Primary 2
Egypt	Year 2
Hong Kong	Primary 2
Indian Subcontinent ¹	Class 2
Indonesia	Year 3
Malaysia	Standard 2
Middle East ²	Class 2
New Zealand/Pacific ³	Year 3
Singapore	Primary 1
Southern Africa ⁴	Grade 2

¹ Indian Subcontinent Region: India, Sri Lanka, Nepal, Bhutan and Bangladesh.

² Middle East Region: United Arab Emirates, Qatar, Kuwait, Saudi Arabia, Bahrain, Oman, Turkey, Lebanon, Tunisia, Morocco, Libya, Algeria, Jordan and Pakistan.

³ Pacific Region: Vanuatu, Papua New Guinea and Fiji.

⁴ Southern Africa Region: South Africa, Botswana, Lesotho, Swaziland, Zimbabwe and Namibia.

INTRODUCTORY
PAPER



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